

2.4 GHz Hairpin Bandpass Filter Design

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Abstract—This report presents the design and electromagnetic (EM) simulation of a 5-pole hairpin bandpass filter for the 2.4 GHz ISM band on standard FR4 substrate ($\epsilon_r = 4.4$, $h = 1.57$ mm). Using Sonnet planar EM simulation, the filter achieves a centre frequency of 2.455 GHz, approximately 200 MHz 3-dB bandwidth, 5.9 dB peak insertion loss, and 27.6 dB return loss at the passband centre, with upper stopband rejection exceeding 40 dB at 2.7 GHz. The design is ready for PCB fabrication with a minimum feature size of 0.2 mm.

Index Terms—bandpass filter, coupled resonators, FR4, hairpin filter, ISM band, microstrip, planar EM simulation, 2.4 GHz.

I. FILTER SPECIFICATIONS

The target filter specifications are summarised in Table I.

TABLE I
TARGET FILTER SPECIFICATIONS

Parameter	Value	Notes
Centre frequency	2.4 GHz	ISM band
Filter type	Chebyshev	0.5 dB in-band ripple
Filter order	5 poles	Steep roll-off
Fractional BW	10%	240 MHz absolute
Substrate	FR4	$\epsilon_r = 4.4$, $\tan \delta \approx 0.02$
Substrate thickness	1.57 mm	Standard two-layer PCB

II. FILTER LAYOUT AND GEOMETRY

The hairpin filter consists of five coupled quarter-wave resonators arranged in a row, each folded into a U-shape. Tapped feed lines at the two outer resonators connect the filter to the 50 Ω system. Fig. 1 shows the Sonnet EM layout of the final design.

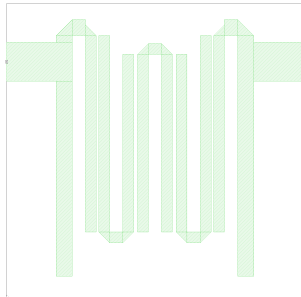


Fig. 1. Sonnet EM layout of the 5-pole hairpin bandpass filter. Five quarter-wave resonators are coupled via edge-proximity gaps. Tapped 50 Ω feed lines connect at ports 1 and 2.

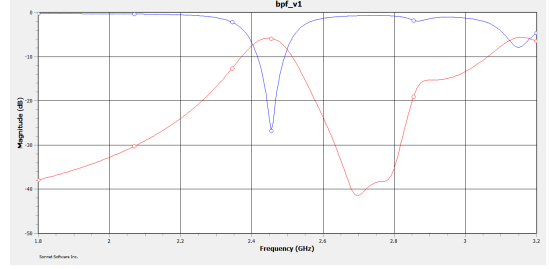


Fig. 2. Simulated S-parameters, 1.8 to 3.2 GHz. $|S_{21}|$ (red) peaks at -5.9 dB at 2.455 GHz. $|S_{11}|$ (blue) reaches -27.6 dB at the passband centre.

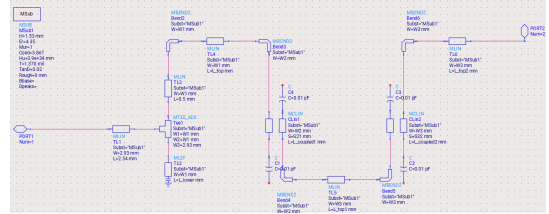


Fig. 3. Final Design of the filter. Simulated results are derived from this data.

III. EM SIMULATION RESULTS

TABLE II
SIMULATED PERFORMANCE SUMMARY

Parameter	Simulated	Target
Centre frequency	2.455 GHz	2.4 GHz
3-dB bandwidth	≈ 200 MHz	240 MHz
Fractional bandwidth	$\approx 8.2\%$	10%
Peak insertion loss ($ S_{21} $)	5.9 dB	< 6 dB (FR4 limit)
Return loss at f_0 ($ S_{11} $)	27.6 dB	> 15 dB
Stopband rejection at 2.7 GHz	> 40 dB	> 30 dB
Filter poles	5	5

IV. CONCLUSION

The design is cleared for PCB fabrication, with a minimum feature size of 0.1 mm on the outer coupling gaps. Post-fabrication VNA measurement will confirm simulation accuracy and determine whether a second layout iteration is needed to correct centre frequency or bandwidth.

REFERENCES

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